

A12 Docs RAG On Premise

Build a RAG pipeline with Milvus

AI VN Team

5.2026

Agenda

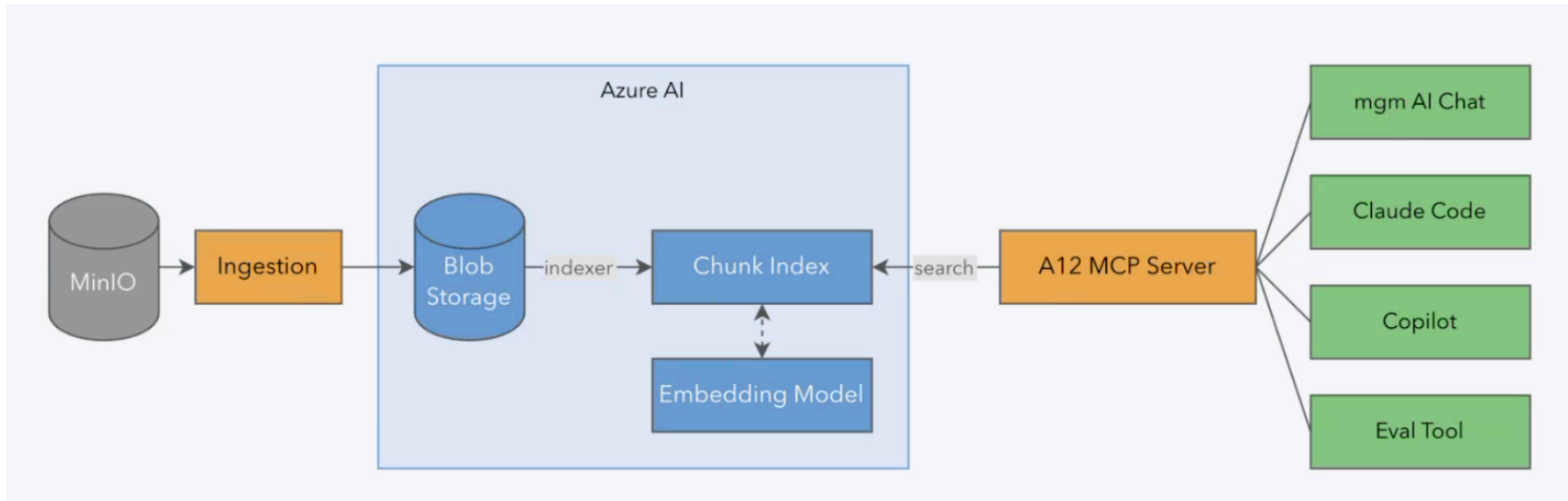
- Goals of moving from Azure cloud to on premise
- Compare architecture solution on Azure cloud and on premise with Milvus
- How to sync data in ingestion on premise
- Compare their chunking strategy
- Compare their retrieval strategy
- Compare their performance
- Leverage relationship of A12 components to help LLM reasoning better

Goals of moving from Azure cloud to on premise

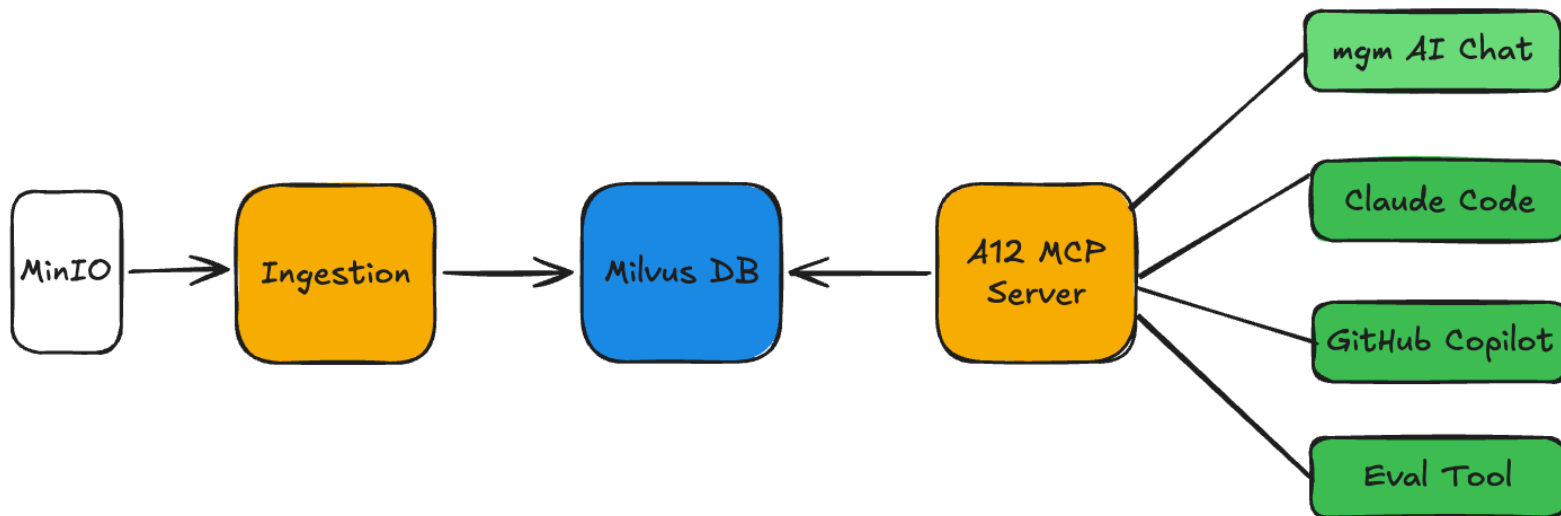
- Minimize dependence on Azure cloud and reduce costs
- Ensure GDPR compliance and data sovereignty on our own infrastructure
- Maintain the same retrieval performance as A12 RAG on Azure
- Make it easy to control and address nuanced problems in A12 RAG

Overview A12 RAG MCP architecture on Azure cloud

- Azure AI search

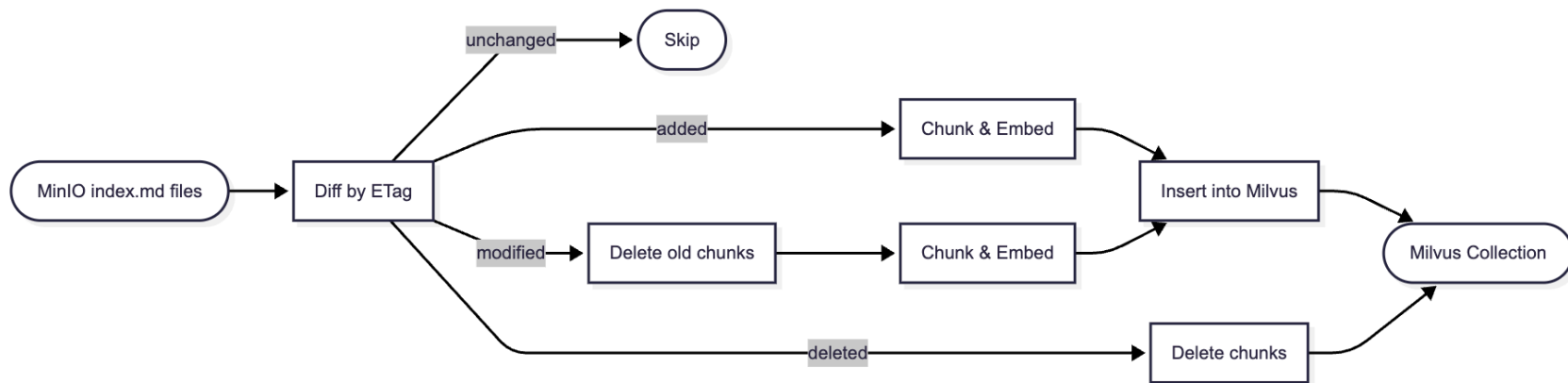


Overview A12 RAG MCP architecture on premise



- Replace Azure AI Search with Milvus
- Remove Azure Storage dependency
- Keep Azure OpenAI for text-embedding-3-large

A12 Docs ingestion on premise (1)



- Ingestion still runs on a schedule
- Milvus manages data sync from MinIO
- Each A12 doc version maps to a partition in the Milvus collection — ANN search is scoped to one or more partitions explicitly

A12 Docs ingestion on premise (2)

- Sample logs during data sync

```
Partitions: 2023_06_ext11
Total chunks: 3729
To sync: 2025.06/ext4/*
```

```
To add 112
To modify 0
To delete 0
Unchanged 0
```

```
Version 2025.06-ext4 (0 deleted 0 modified 112 added)
```

```
[2/112] 2025.06/ext4/utils_logging_collections/utils-logging-documentation/index.md → 18 chunks
[3/112] 2025.06/ext4/cms/cms-dev-docs/index.md → 21 chunks
[4/112] 2025.06/ext4/utils_logging_collections/utils-collections-documentation/index.md → 7 chunks
[1/112] 2025.06/ext4/base/base-documentation-bundle/index.md → 16 chunks
```

A12 Docs ingestion on premise (3)

default (3)

a12_docs (3,729)

a12_sync_jobs (1)

a12_sync_state (58)

Collections Properties

+ Collection

+ Import File

Refresh

Search by name

☐	Name	Status	Approx Entity Count [?]	Description	Alias [?]	Created At	Updated At
☐	a12_docs	Loaded	3,729	--	-- +	5/4/2026, 4:29:19 PM	5/4/2026, 4:29:19 PM
☐	a12_sync_jobs	Loaded	1	--	-- +	5/4/2026, 4:29:21 PM	5/4/2026, 4:29:21 PM
☐	a12_sync_state	Loaded	58	--	-- +	5/4/2026, 4:29:20 PM	5/4/2026, 4:29:20 PM

- a12_docs : stores document chunks
- a12_sync_jobs : tracks ingestion jobs
- a12_sync_state : tracks synced document files

Chunking Challenges in A12 Docs (1)

- **Problem 1:** A12 Docs frequently contain intra-references — links pointing to other sections within the same document

2023.06-ext11
🔍

- Get Started
- Tutorials
- Components**
 - Base
 - Client
 - Data Distribution
 - Data Services**
 - Engines
 - Expressions
 - Kernel
 - Model Editor (SME)
 - Notification Center
 - UAA

Data Services

The most common use-cases follow:

Single Server

If you run just a single Data Services server, you can use embedded Active MQ broker to manage the queue. To achieve this, use the Asynchronous CDD Spring profile together with the Embedded JMS Spring profile.

This is primarily intended for development and testing environments, and it is also possible to run CDD processing synchronously. Please see the Synchronous CDD Spring profile and pay attention to the Synchronous CDD Processing section below.



Keep in mind, that synchronous processing could be a performance bottleneck in a production environment.

Chunking Challenges in A12 Docs (2)

- The RAG pipeline cannot answer questions that depend on content from these intra-references

Question Details

What is required to enable CDD processing on a single server?

dataservices

easy

Evaluation Metrics

Answer Relevance ⓘ

0.643

Answer Correctness ⓘ

0.548

Context Precision ⓘ

0.589

Context Recall ⓘ

0.000

Expected Answer

For a single server setup, an embedded Active MQ broker can be used with the following configurations: `spring.artemis.mode=embedded`, `spring.artemis.embedded.enabled=true`, and enabling the producer and consumer with appropriate properties.

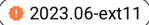
Generated Answer

To enable CDD processing on a single server, you need to:

1. Use the **Asynchronous CDD** Spring profile (`spring-profile-cdd\_async`) along with the **Embedded JMS** Spring profile (`spring-profile-embedded\_jms`) to manage the dirty-documents queue with an embedded ActiveMQ broker.
2. Set the following configuration properties:
 - `mgmtp.a12.dataservices.cdd.dirtyDocuments.async.consumer.enabled=true`
 - `mgmtp.a12.dataservices.cdd.dirtyDocuments.async.producer.enabled=true`

Chunking Challenges in A12 Docs (3)

- **Problem 2:** A12 Docs contain long code blocks that exceed chunk size limits




Helm A12 Stack

This is values.yaml example. You must change 127-0-0-1 to the IP of your cluster, and storageClass to the storage class of your cluster.

```

1 a12-secrets:
2   dockerConfigJson:
3     enabled: true
4     username: admin
5
6   postgresql:
7     enabled: true
8   dataServices:
9     enabled: true
10  camunda:
11    enabled: true
12  contentStore:
13    enabled: true
14
15  hazelcastManagementCenter:
16    enabled: true
17    username: admin
18
19  jwt:
20    enabled: true
21
22  workflows:
23    restClient:
24      enabled: true
25      username: admin

```

Chunking Challenges in A12 Docs (4)

- **Problem 3:** A12 Docs contain long table blocks that span multiple chunks

2023.06-ext11

Workflows

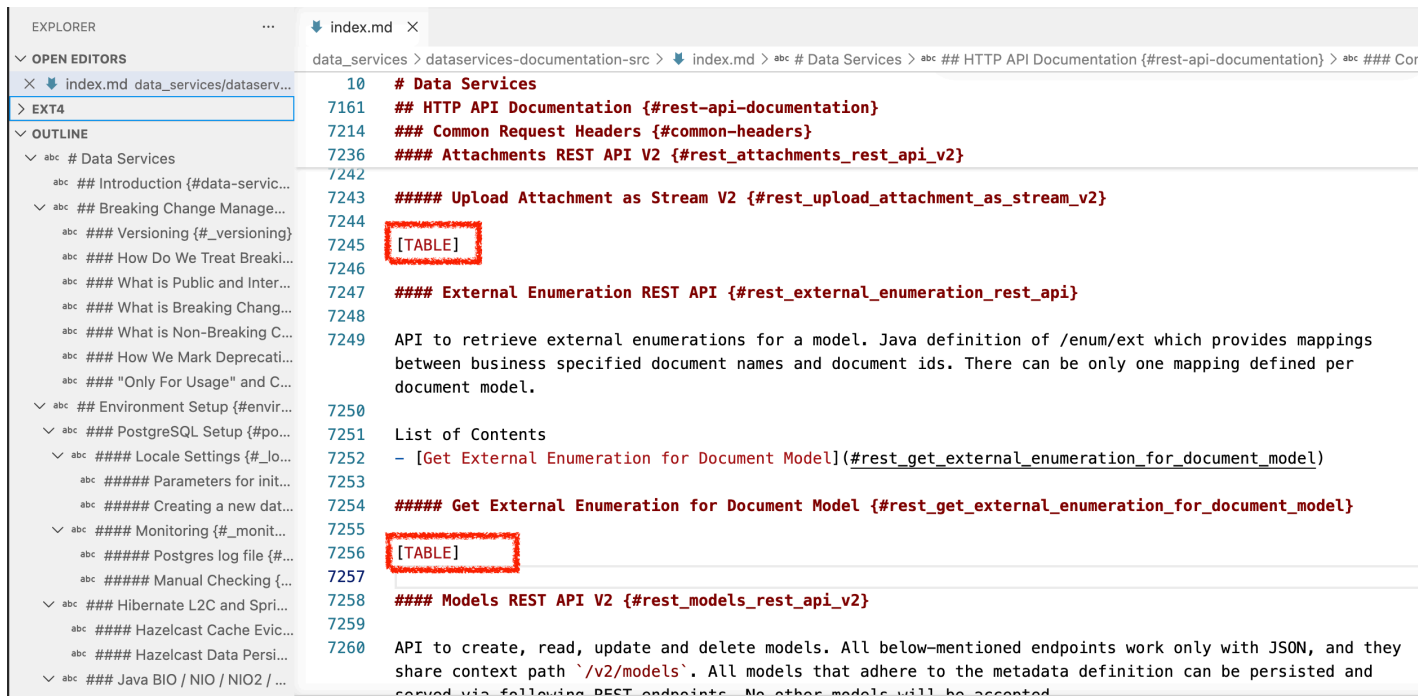
Overall
Modeling
Development
Get Started
Tutorials
Components
Base
Client
Data Distribution
Data Services
Engines
Expressions
Kernel
Model Editor (SME)
Notification Center
UAA
Utils
Widgets
Workflows

Scopes for Process-Related Authorization Rules

Endpoint	Arguments available in authorization file	Scope
Endpoint to start the process: <code>/api/process-definition/{key}/start</code>	<code>#processDefinitionKey:</code> String <code>#inputDocuments:</code> List<InputDocumentDT0>	ProcessDefinitionCreateInst: - runs before entering the method
Endpoint to get list of process definitions: <code>/api/process-definition</code>	-	ProcessDefinitionRead - runs before entering the method
	<code>#resource:</code> ProcessDefinition	ProcessDefinitionFilter - runs after method execution and allows filter list of process definitions, by applying defined rules to each elen in the list
Endpoint to get process diagram: <code>/api/process-definition/{key}/diagram</code>	<code>#processDefinitionKey:</code> String	ProcessDefinitionRead - runs before entering the method

Chunking Challenges in A12 Docs (5)

- **Problem 4:** A12 Docs contain table placeholders in Markdown that are replaced during AsciiDoctor builds



```

EXPLORER
└─ OPEN EDITORS
  └─ index.md data_services/dataserv...
    └─ EXT4
      └─ OUTLINE
        └─ abc # Data Services
          └─ abc ## Introduction {#data-serv...
            └─ abc ## Breaking Change Manage...
              └─ abc ### Versioning {#_versioning}
                └─ abc ### How Do We Treat Breaki...
                  └─ abc ### What is Public and Inter...
                    └─ abc ### What is Breaking Chang...
                      └─ abc ### What is Non-Breaking C...
                        └─ abc ### How We Mark Deprecati...
                          └─ abc ### "Only For Usage" and C...
                            └─ abc ## Environment Setup {#envir...
                              └─ abc ### PostgreSQL Setup {#po...
                                └─ abc ##### Locale Settings {#_lo...
                                  └─ abc ##### Parameters for init...
                                    └─ abc ##### Creating a new dat...
                                      └─ abc ##### Monitoring {#_monit...
                                        └─ abc ##### Postgres log file {#...
                                          └─ abc ##### Manual Checking {...
                                            └─ abc ### Hibernate L2C and Spr...
                                              └─ abc ##### Hazelcast Cache Evic...
                                                └─ abc ##### Hazelcast Data Persi...
                                                  └─ abc ### Java BIO / NIO / NIO2 / ...

data_services > dataservices-documentation-src > index.md > abc # Data Services > abc ## HTTP API Documentation {#rest-api-documentation} > abc ### Cor
10 # Data Services
7161 ## HTTP API Documentation {#rest-api-documentation}
7214 ### Common Request Headers {#common-headers}
7236 #### Attachments REST API V2 {#rest_attachments_rest_api_v2}
7242
7243 ##### Upload Attachment as Stream V2 {#rest_upload_attachment_as_stream_v2}
7244
7245 [TABLE]
7246
7247 ##### External Enumeration REST API {#rest_external_enumeration_rest_api}
7248
7249 API to retrieve external enumerations for a model. Java definition of /enum/ext which provides mappings
between business specified document names and document ids. There can be only one mapping defined per
document model.
7250
7251 List of Contents
7252 - [Get External Enumeration for Document Model] {#rest_get_external_enumeration_for_document_model}
7253
7254 ##### Get External Enumeration for Document Model {#rest_get_external_enumeration_for_document_model}
7255
7256 [TABLE]
7257
7258 ##### Models REST API V2 {#rest_models_rest_api_v2}
7259
7260 API to create, read, update and delete models. All below-mentioned endpoints work only with JSON, and they
share context path `/v2/models`. All models that adhere to the metadata definition can be persisted and
saved via following REST endpoints. No other models will be accepted

```

Chunking Challenges: New features in roadmap

- Assets [PRIO 1]
 - Statistic: What assets exist?
 - Add general pre-process functionality between download from MinIO and Upload to Azure Blob.
 - First preprocess: Move link processing from retrieval to ingestion
 - Inline tables [PRIO 0]
 - Inline short code snippets
- Observability: Trinh <https://jira.mgm-tp.com/jira/browse/DS-2027?jql=component%20%3D%20a12-rag>
- Publish Docker Image to Artifactory [PRIO 1]

Find Concept

- **Prepare for external usage [PRIO 0, June Release]**
 - Prepare for public vs enterprise source bucket (see <https://jira.mgm-tp.com/jira/browse/A12DOC-1057>)
 - Auth (GetA12 Identity Token?)  [DS-2064](#) - a12-rag-mcp: Authenticate A12 RAG MCP Server **IN PROGRESS**
 - Security Review (SP, Benjamin Weller)
- Code Example MCP Tool [PRIO 2]
 - Find: high-quality code snippets. Source Ideas:
 - Showcases
 - A12 Source Code + Test Codee
 - A12 Gems
- Model Example MCP Tool [PRIO 2]
- Assets (Images, HTML, Code) [PRIO 3]
 - Large Code snippets? (Code more difficult than prose)
 - Transcribe Images (Azure) and inline (with link to img)
- Move MCP Deployment to C12 Cluster [PRIO 2]

Compare their chunking strategy - Common steps

AZURE AI SEARCH

Split by headings (h1-h3)

Chunk by character limit (2000 chars, 500 overlap)

MILVUS SEARCH

Split by headings (h1-h4)

Chunk by recursive splitter, cap = 3000 chars, no overlap

Compare their chunking strategy - Specific steps

AZURE AI SEARCH	MILVUS SEARCH	PURPOSE
None	Append breadcrumb to chunk text	Separates similar text across different components; anchors chunks to their source document context
None	Detect inline code and tables	Future: preserve table headers across chunks, help the LLM retrieve full blocks via tools
None	Detect intra-references in chunks	Future: return referenced content within the chunk
None	Detect sections spanning multiple chunks	Provide a tool to retrieve the full section

Compare their chunking strategy - Chunk text in Milvus

default (3) / a12_docs

default (3)

- a12_docs (14,858)
- a12_sync_jobs (5)
- a12_sync_state (267)

Schema Data Vector Search Partitions Segments

Query Expression

Please enter your query expression, eg: id > 0

1 selected Import File Insert Sample Data Empty

is_overflow	section_id
UE	5d01efa4-ee9e-461a-b9a2-08d1c7e693fa
LSE	a7c8cb46-1fcd-4636-99ff-21e14889232a
LSE	9055fd8f-af10-437e-8031-37a842dbebeb
LSE	6ee9eb95-7ca1-42fb-a8a9-272d184e1005
UE	d5798bdb-9ed0-413c-9d6e-4e2324c4361e
UE	e95e0a2e-6266-4ae2-85b2-183593d07d0c
UE	1662836e-9172-4c8c-b535-15459f513baa
UE	023ed750-ef6d-42ec-a7ee-15f65727b235
LSE	5a07f2cb-b0b8-4b45-98c3-1a3bbd14ef5c
LSE	51aaf994-e1b4-4438-b11f-5a89e03bc597

Entity Data

parent_id

61a58d23-1ee1-4d5f-8a8c-3ad2209ba37a

text

A12 Data Services > [](#json-rpc-endpoint){.anchor}JSON-RPC Endpoint > JSON-RPC 2.0 Core Operations > ADD_LINK

ADD_LINK

- This operation does not ensure the deferred data integrity constraints (upper limit). Instead, the ADD LINK operation fires events that fill a ThreadLocal collection, which is used by 'RelationshipLinkOperationValidator' only after all the operations are finished. Calling this operation directly therefore might corrupt link integrity and cause issues with subsequent RPC operations due to the usage of ThreadLocal collections for validation.

Call sequence {#_call_sequence_14}

![link addLinkOperationSequence]
(https://geta12.com/docs/2023.06/ext11/data_services/dataservices-documentation-src/assets/images/link_addLinkOperationSequence.svg)

Figure 14. Sequence of ADD_LINK calls.

```

| | |
|---|---|
| | This sequence is compliant with the default settings without caches and with rpc
enabled which should show most of the events and security checks. Using different

```

Enhance chunking strategy - Consider aspects

ASPECT	AZURE AI SEARCH	ON PREMISE
Extend chunking logic	Difficult — requires additional Azure services (e.g. Azure Functions)	Easy — just change the code
Cost	Higher cost	Free
Dependencies	More dependencies	No additional dependencies

Compare retrieval strategy

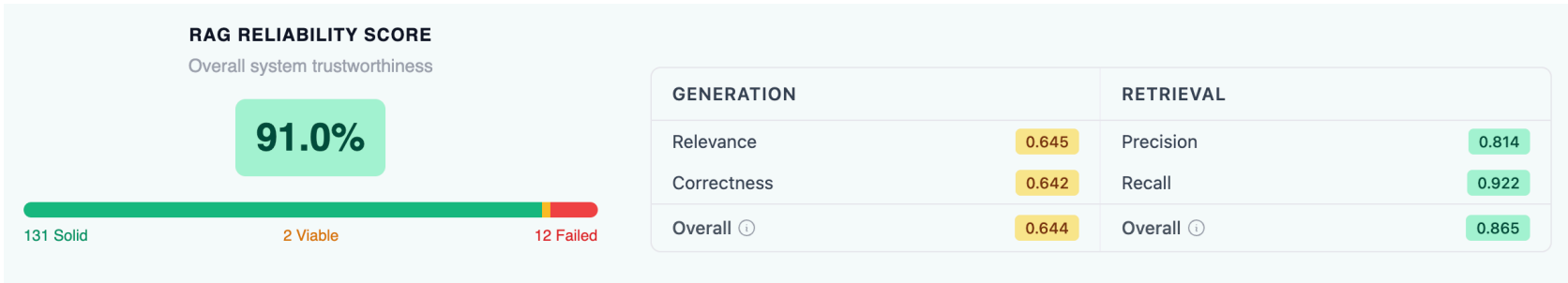
AZURE AI SEARCH	MILVUS SEARCH
Hybrid search	Hybrid search
RRF	RRF
Built-in semantic reranker	None

Compare retrieval strategy — Reranker flexibility

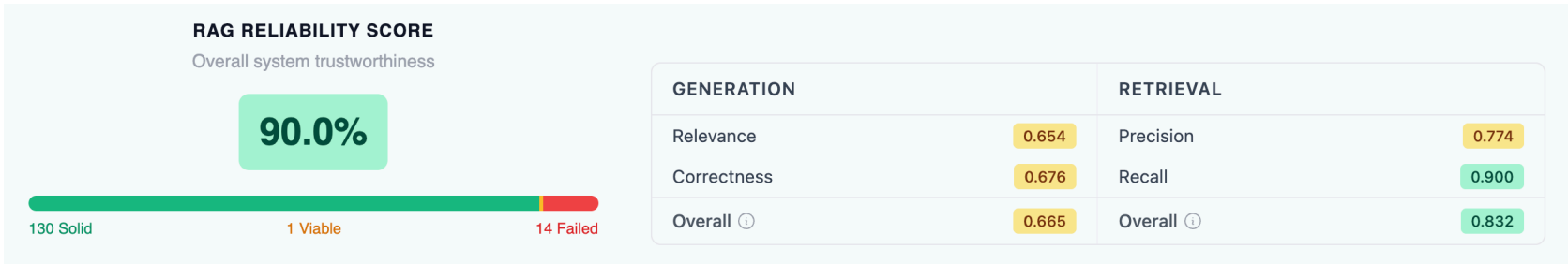
- **Azure AI Search:** the semantic reranker is built-in and optional — no alternative rerankers are available
- **On-premise:** choose from Milvus's built-in rerankers or implement a fully custom reranker

Compare A12 RAG Evaluation

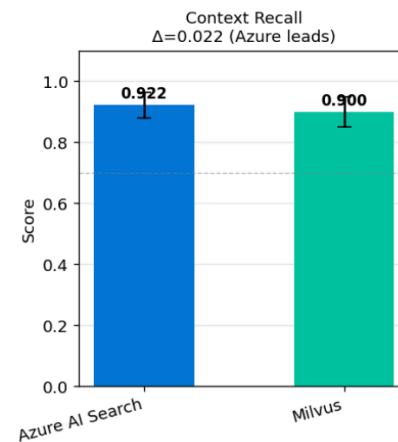
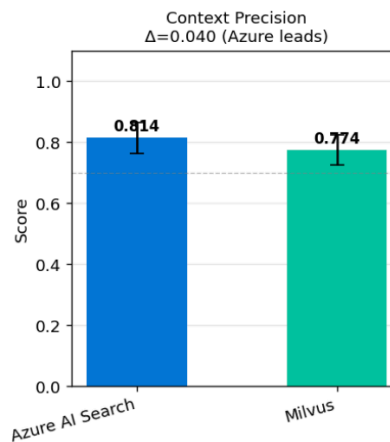
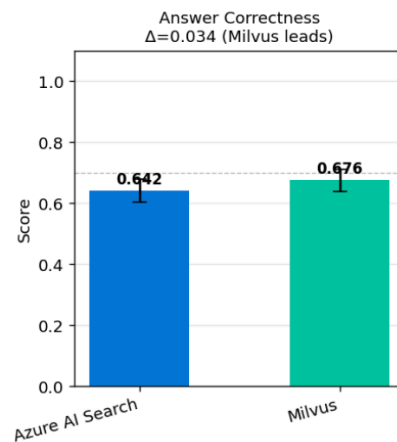
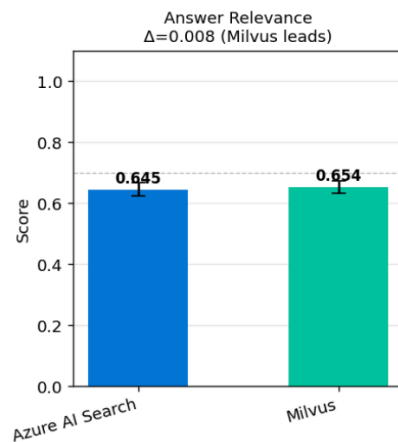
- Azure AI Search |



- Milvus search



Evaluation Insights between Azure AI Search and Milvus search



Leveraging A12 component relationships to improve search

- A12 docs are technical documents with strong dependencies between components.

LAYER	NAME	COMPONENTS
0	Utilities & Foundation	utils_localization, utils_logging_collections, utils_server_connector, base, plasma
1	Core Modeling Foundation	kernel, expression
2	Design-Time (Modeling)	sme, diagram_editor, tdg
3	Server Runtime	data_services, uaa, user_management, transformer, data_distribution, workflows
4	Client Runtime	client, form_engine, overview_engine, tree_engine, crud, relationship_engine, content_engine, notification_center
5	UI & Output	widgets, print_engine, cms
—	Cross-cutting	build_and_deployment, project_template, overall

Leveraging A12 component relationships (1)

- Enhanced MCP instructions to improve LLM reasoning

MCP_INSTRUCTIONS:

```
'Use search_a12_docs to search the A12 knowledge base (geta12.com). ' +
'A12 is a model-driven enterprise application platform with ~28 components organized in 6 architectu
'plus cross-cutting components:\n' +
'- Layer 0 – Utilities & Foundation: utils_localization, utils_logging_collections, utils_server_con
'- Layer 1 – Core Modeling Foundation: kernel, expression\n' +
'- Layer 2 – Design-Time (Modeling): sme, diagram_editor, tdg\n' +
'- Layer 3 – Server Runtime: data_services, uaa, user_management, transformer, data_distribution, wo
'- Layer 4 – Client Runtime: client, form_engine, overview_engine, tree_engine, crud, relationship_e
'- Layer 5 – UI & Output: widgets, print_engine, cms\n' +
'- Cross-cutting – Lifecycle & Onboarding: build_and_deployment, project_template, overall',
```


Leveraging A12 component relationships (2)

- Enhanced search tool descriptions so the LLM can reason about results and refine follow-up searches automatically

MCP_TOOL_DESCRIPTION:

```
'Search A12 platform documentation via semantic search. Returns ranked document chunks with scores. '
'Supports filtering by A12 version and component. Does not generate answers. Returns raw documentatio
'## Component Search Strategy\n\n' +
'Most questions span multiple A12 components. If the first search surfaces a component you weren\'t '
'expecting, use the cluster list below to pick 1-2 related components for follow-up searches via the
'\`a12_component` parameter.\n\n' +
'Clusters are intent-based, not exclusive – a component may appear in multiple clusters depending on
'Key clusters to search together:\n' +
'- Frontend (UI rendering questions): client, form_engine, overview_engine, widgets\n' +
'- Modeling pipeline (how-to-model questions): sme, kernel, form_engine\n' +
'- Backend/API (server logic): data_services, uaa, workflows\n' +
'- Workflow (BPMN tasks): workflows, form_engine, data_services, client\n' +
'- Notifications (alerts/messaging): data_distribution, notification_center\n' +
'- Deployment (infra/setup): build_and_deployment, project_template, uaa, data_services\n' +
'- Getting started (orientation): overall, project_template',
```

Patrick Schmidt, 2 months ago

Claude Code with A12 RAG MCP on premise



Claude Code with A12 RAG MCP in Azure



End of Briefing

Questions and discussion